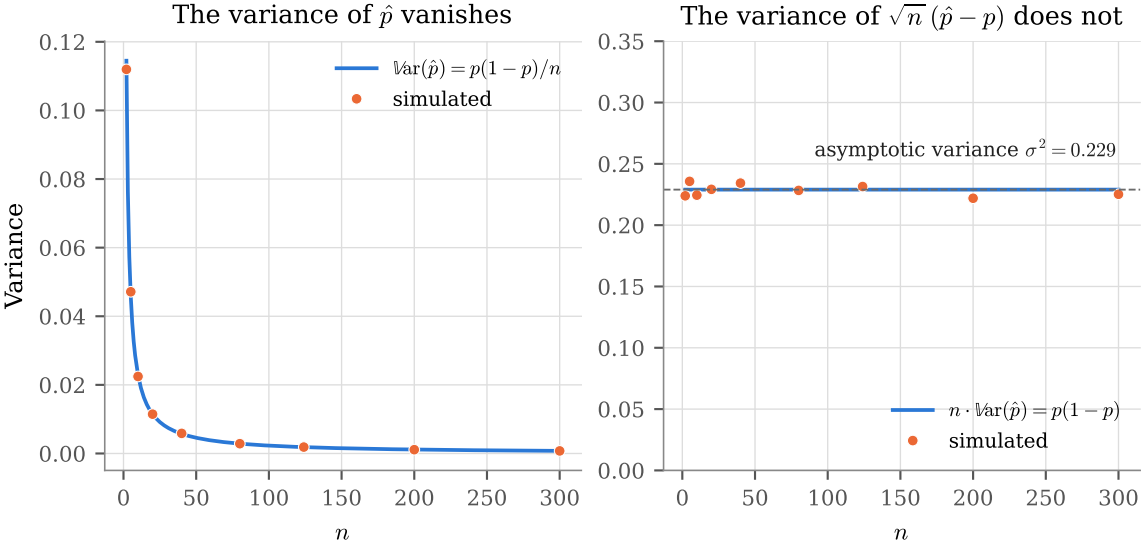
The variance of $\hat{p}$ vanishes
$\mathbb{V}\mathrm{ar}(\hat{p}) = p(1-p)/n$
simulated
Variance
0.12
0.10
0.08
0.06
0.04
0.02
0.00
0
50
100
150
200
250
300
n
The variance of $\sqrt{n}\,(\hat{p}-p)$ does not
asymptotic variance $\sigma^2 = 0.229$
$n \cdot \mathbb{V}\mathrm{ar}(\hat{p}) = p(1-p)$
simulated
0.35
0.30
0.25
0.20
0.15
0.10
0.05
0.00
0
50
100
150
200
250
300
n